

PG Installation Analysis

Software User Guide OrcaFlex Automation Tool (Reel-lay) 2nd End Structure Laydown

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VERSION RECORD SHEET

Version	Issue Date	Software Version	Section	Description of Update
00	24.10.2025	0.1.0	All	-

TOOL REVISION HISTORY

Version	Issue Date	Description of Update
0.1.0	24.10.2025	First Draft

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1. INTRODUCTION

The analysis requirements for the 2nd end structure laydown are as follows:

- Static step-by-step Laydown tables,
- Dynamic screening of the steps during the Laydown sequence,
- Allowable sea-states and limiting parameters.

1.1 SOFTWARE OVERVIEW

This tool is built to automate the following processes:

- Create 2nd end structure laydown OrcaFlex models,
- Create initial buoyancy module deployment steps,
- Lower the 2nd end structure by adjusting wire payout, vessel position, and ramp angle,
- Create static laytables for the Laydown sequence,
- Perform 200s peak runs for dynamic screening to find out critical steps,
- Find allowable sea-states and limiting parameters.

1.2 SYSTEM REQUIREMENTS

OrcaFlex 11.3 and Python 3.8 (or above).

1.3 SETUP

Download the application file (folder called 'main') and run the main.exe application.

1.4 DOCUMENT PURPOSE

The purpose of this document is to provide instructions for installation, running of the Laydown automation tool as well as information regarding its capabilities and limitations.

1.5 DEFINITIONS

A&R	Abandonment and Recovery
CRA	Corrosion Resistant Alloy
GUI	Graphical User Interface
UI	User Interface
WD	Water Depth

1.6 REFERENCES

1. DNV-ST-F101 Submarine Pipeline Systems (2017)
2. PG-ENG-ST-010 Installation Analysis Guidelines for Reel Lay
3. PG-ENG-ST-007 PG Practice – Reeled Pipelay Lay Tables

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2. PRODUCT FEATURES

The generalized process of the tool is summarised as follows:

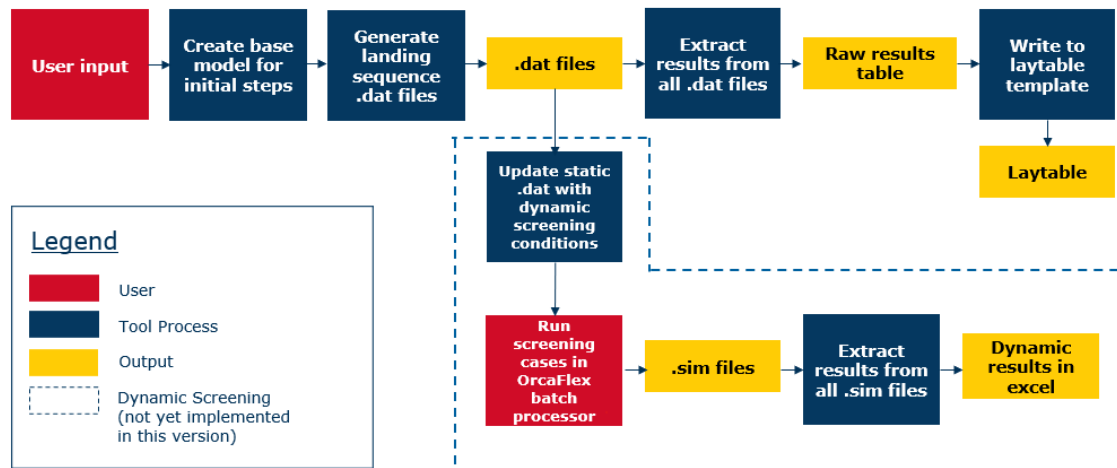


Figure 2-1 - 2nd End Laydown Tool Process

2.1 CREATE BASE MODEL

Given the environment, pipe, vessel, structure, and buoyancy module data, the tool will create OrcaFlex base models. The first stage of the process is to create the base models for the following initial steps:

1. When the structure is first fully submerged prior to buoyancy module activation,
2. When the first buoyancy module is activated,
3. When the second buoyancy module, if any, is activated, and so on until all buoyancy modules have been activated.

2.2 GENERATE SEQUENCE

The wire is paid out and the vessel is moved forward until the structure is deemed to have landed on the seabed. The amount of wire paid out is determined by the user by the input *Max Payout Rate*, unless a step at a specific water depth is required (detailed below). The vessel movement is adjusted accordingly to minimize wire lift at the vessel ramp.

The tool aims to capture the following specific steps, in addition to those mentioned in Section 2.1:

1. When the structure is lowered to a water depth where it is deemed that its hydrodynamic properties can switch from *shallow* to *deep* water properties, in this case 100m WD,
2. When the structure (if modelling a winged mudmat) achieves a water depth at which the mudmat wings are opened,
3. When the structure is at 30m, 20m, 10m, and 5m clearance from the seabed,

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4. When the pipe's integrity limits (DNV LCC, von Mises stress, strain) exceeds a limit. To mitigate this, the programme jacks down the vessel ramp whilst paying out 0 wire.

2.3 STATIC RESULTS AND LAYTABLES

The static result of the laydown sequence are written to two (2) Excel spreadsheets:

1. A raw results table containing the necessary analysis outputs,
2. A laytable as per the set PG standard.

2.4 DYNAMIC SCREENING

[HOLD]

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3. GUIDE TO RUNNING A CALCULATION

3.1 STEP 1: RUN THE MAIN APPLICATION.

Double click on the *Main* application to launch the tool.

☐ Name	Date modified	Type
_internal	24/10/2025 08:53	File folder
vessel_database	24/10/2025 08:54	File folder
main	24/10/2025 08:53	Application
Laytable_Template	23/10/2025 15:06	Microsoft Excel Worksheet
infoWindow.ui	16/10/2025 14:01	UI File
LDWindow.ui	23/10/2025 14:41	UI File

Figure 3-1 - Starting the application.

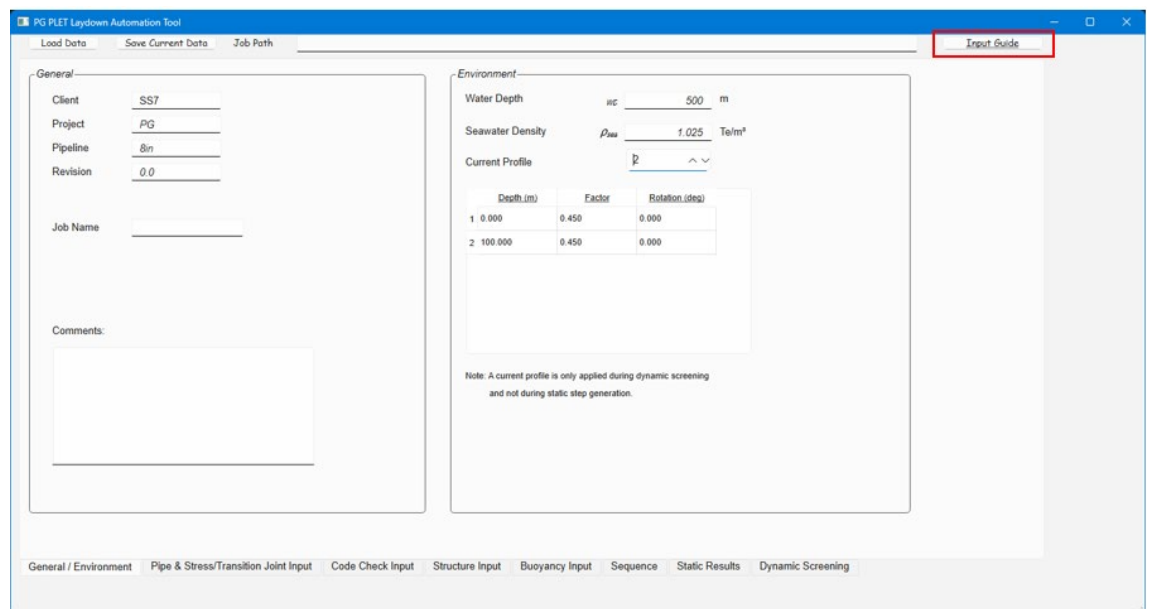


Figure 3-2 - Tool User Interface

Note that this User Guide can be accessed via the button highlighted in red in Figure 3-2.

3.2 STEP 2: START POPULATING THE INPUTS

The user input is divided into eight (8) tabs, which are:

1. General/Environment
2. Pipe Input
3. Code Check Input
4. Structure Input
5. Buoyancy Input
6. Sequence
7. Static Results

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8. Dynamic Screening [HOLD]

3.2.1 Tab 1 – General/Environment Input

In the General tab, the following inputs can be made:

- Client
- Project
- Pipeline
- Revision
- Job Name
- Comments

The following Environmental inputs can also be made in this tab:

- Water depth [m]
- Seawater density [Te/m³]
- Current profile

Note that the current profile is not applicable for static analyses.

3.2.2 Tab 2 – Pipe Inputs

In this tab, the following properties are defined:

For the main pipe:

- Linetype name
- Pipeline type (Single Pipe or Pipe-In-Pipe [HOLD])
- CRA liner usage
- OD, WT [mm]
- Material density [Te/m³]
- Coating thickness [mm]
- Coating density [Te/m³]
- Young's modulus, E [MPa]
- Added mass coefficient, Cax
- Drag coefficient, Cdx
- Seabed normal and axial friction coefficients
- Stress-strain relationship (Linear or Ramberg-Osgood)
- Pipe contents (empty, free-flooding, or flooded)

For the stress joint (if called for):

- Effective length [m]
- WT, ID [mm]
- Transition joint length [m]

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PG PLET Laydown Automation Tool

Load Data Save Current Data Job Path Input Guide

Line Type Name: SS7 pipeline

Pipeline Type: Single Pipe

CRA Lined Pipe?: ☐

Outer Diameter: 323.9 mm

Inner Diameter: 273.9 mm

Wall Thickness: 25 mm

Material Density: 7.85 T/m³

Coating Thickness: 2 mm

Coating Density: 0.85 T/m³

Young's Modulus E: 212000 MPa

Added Mass Coeff. (Cax): 1.0 -

Drag Coeff. (Cdx): 1.2 -

Seabed Normal Friction Coeff.: 0.5 -

Seabed Axial Friction Coeff.: 0.4 -

Pipe Contents Data

Pipeline Contents: Flooded

Contents Density: 1.025 T/m³

Stress Joint Necessary? ☒

Stress Joint

Effective Length of Stress Joints: 4.5 m

Stress Joint OD: 460.0 mm

Stress Joint WT: 30 mm

Stress Joint ID: 400 mm

Transition Joint Data

Effective Length of Transition Joint: 60 m

Stress-Strain Relationship: Ramberg-Osgood

Stress-Strain Model: X65 - R-O

Reference Stress: 418.13 MPa

Parameter K: 0.00087

Parameter n: 17.3

General / Environment Pipe & Stress/Transition Joint Input Code Check Input Structure Input Buoyancy Input Sequence Static Results Dynamic Screening

Figure 3-3 - Pipe Inputs Tab

3.2.3 Tab 3 – Code Check Input

The user can input values for the following code checks:

- DNV-ST-F101
- API RP 1111 BEP

PG PLET Laydown Automation Tool

Load Data Save Current Data Job Path Input Guide

Combined Loading Criteria - Load Controlled Condition

Design Data:

Pipe Data:

Outer Diameter: D_o 323.9 mm

Nominal Thickness: t_{nom} 25.0 mm

CRA Thickness: Include? ☐

Fabrication thickness tolerance: t_{fab} 0 mm

Corrosion Allowance: t_{corr} 0 mm

Ovality (DNV Definition): f_o 0.03 %

Pressure Data:

Minimum Internal Pressure: p_{min} 0 MPa

Design Depth: w_d 1001 m

API RP 1111 Factors:

Allowable Load Factor: F_a 0.96 -

Collapse Factor: F_c 0.70 -

Bending Safety Factor: F_{bs} 2.00 -

Strain Amplification Factor: SAF 1.0 -

Stress Joint Data:

Outer Diameter: D_{sj} 460.0 mm

Nominal Thickness: t_{sj} 30.0 mm

Material Data:

Material Grade: DNV SMLS 450 PD

Young's Modulus: E 212000 MPa

Poisson's Ratio: ν 0.3 -

Specified Min. Yield Stress: SMYS 450 MPa

Specified Min. Tensile Stress: SMTS 535 MPa

Specified Min. Yield Stress (CRA): $SMYS_{CRA}$ 0 MPa

Specified Min. Tensile Stress (CRA): $SMTS_{CRA}$ 0 MPa

Fabrication Factor: α_{fab} 0.93 -

Material Strength Factor: α_s 0.96 -

Load Factors:

Functional Load Factor: γ_F Case A 1.2 Case B 1.1 -

Environmental Load Factor: γ_E 0.7 1.3 -

Condition Load Effect Factor: γ_C 1 -

Safety Class Resistance Factor: γ_{sc} 1.04 -

Material Resistance Factor: γ_m 1.15 -

General / Environment Pipe & Stress/Transition Joint Input Code Check Input Structure Input Buoyancy Input Sequence Static Results Dynamic Screening

Figure 3-4 - Code Check Tab

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3.2.4 Tab 4 – Structure Input

The user specifies the following inputs for the 2nd end structure, depending on the selection for the structure modelling:

Frame Only Modelling

Note that specification for CoG (centre of gravity) coordinates assumes that (0, 0, 0) is at the *drawing centre* of the structure.

PLET Data:

Frame:

Length	6.035	m
Width	2.082	m
Height	2.142	m
Weight in Air	22.23	Te
Weight in Water	19.34	Te
CoG X	0.144	m
CoG Y	0.0	m
CoG Z	0.4	m
Perforation X Y Z	0.25 0.25 0.05	-

PLET modelling option: Frame only

General / Environment | Pipe & Stress/Transition Joint Input | Code Check Input | Structure Input | Buoy

Figure 3-5 - Inputs for 'Frame Only' model

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Frame & Mudmat (Without Wings)

If a mudmat is to be modelled (without wings), the inputs are as shown in Figure 3-6. Note that CoG coordinate inputs for the frame and mudmat are based on their respective drawing centre origins.

The connection of the mudmat to the frame is specified based on the offset of the frame to the mudmat, as illustrated in Figure 3-7.

PLET Data:

Frame:

Length	6.035	m
Width	2.082	m
Height	2.142	m
Weight in Air	22.23	Te
Weight in Water	19.34	Te
CoG X	0.144	m
CoG Y	0.0	m
CoG Z	0.4	m
Perforation X Y Z	0.25 0.25 0.05	-

PLET modelling option:
Frame & Mudmat (No Wings) ▾

Mudmat

Length	8.985	m
Width	9.843	m
Height	0.508	m
Weight in Air	12.805	Te
Weight in Water	11.14	Te
CoG X	0.116	m
CoG Y	0	m
CoG Z	0.132	m
Perforation X Y Z	0.03 0.03 0.03	-
Connection to Frame X Y Z	0.402 0.0 -3.11	m

General / Environment | Pipe & Stress/Transition Joint Input | Code Check Input | Structure Input | Buoy

Figure 3-6 - Frame & Mudmat (No Wings)

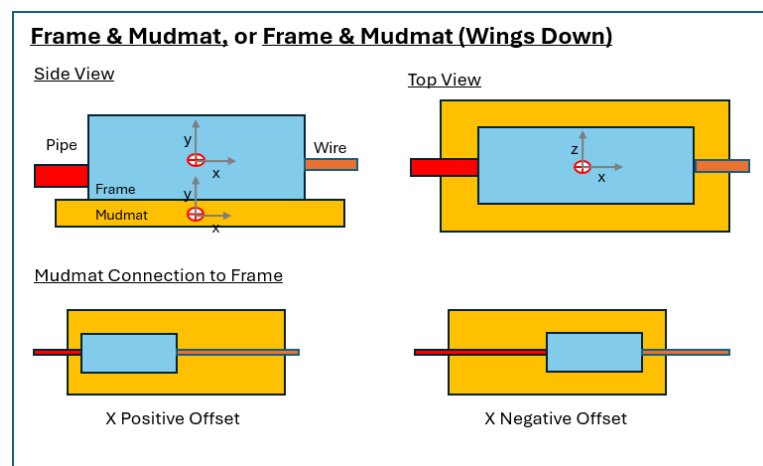


Figure 3-7 - Mudmat Connection Point to the Frame

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Frame & Mudmat (With Wings)

If a mudmat is to be modelled with wings (to be deployed at a certain water depth prior to landing on the seabed), the following inputs are required:

PLET Data:

Frame (with Wings Down):

Length	6.035	m
Width	2.082	m
Height	2.142	m
Weight in Air	22.23	Te
Weight in Water	19.34	Te
CoG X	0.144	m
CoG Y	0.0	m
CoG Z	0.4	m
Perforation X Y Z	0.25 0.25 0.05	-

Mudmat (with Wings Down)

Length	8.985	m
Width	9.843	m
Height	0.508	m
Weight in Air	12.805	Te
Weight in Water	11.14	Te
CoG X	0.116	m
CoG Y	0	m
CoG Z	0.132	m
Perforation X Y Z	0.03 0.03 0.03	-
Connection to Frame X Y Z	0.402 0.0 -3.11	m

PLET modelling option:

Frame & Mudmat (With Wings) ▾

Frame & Mudmat (with Wings Up)

Length	8.985	m
Width	2.997	m
Height	4.083	m
Weight in Air	35.03	Te
Weight in Water	30.48	Te
CoG X	0.389	m
CoG Y	0.0	m
CoG Z	-0.257	m
Perforation X Y Z	0.25 0.25 0.25	-

General / Environment

Pipe & Stress/Transition Joint Input

Code Check Input

Structure Input

Buoy

Figure 3-8 - Frame & Mudmat (With Wings)

Note that in this case, the user will need to specify mudmat parameters for when the wing is up (the first stage) and for when the wing is down (the second stage, nearing landing).

The modelling strategy of the mudmat is clarified in Figure 3-9.

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Laydown Structure Types

TYPE A : Frame only, or Frame & Mudmat modeled as 1 structure (Integrated)

TYPE B : Frame modelled with a Separate Mudmat

Number immediately succeeding the letter denotes hydrodynamic property differences.

1 : Near-Surface

2 : Deep zone

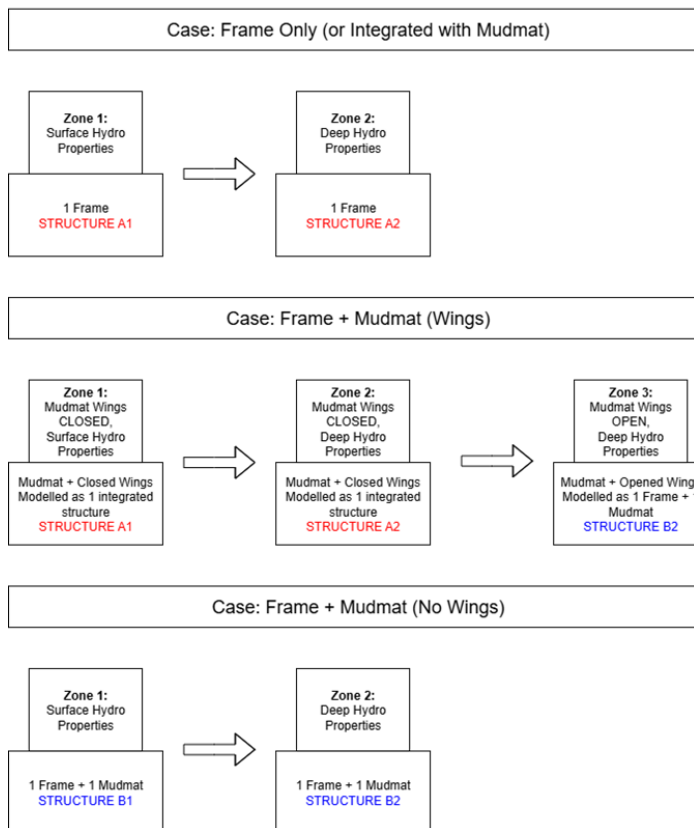


Figure 3-9 - Mudmat Wings Modelling Strategy

A&R Connection

The user is given two (2) options for the A&R wire connection to the structure:

- Padeye connection, where a single connection point is used to attach the wire to the structure, defined as a single coordinate on the frame,
- Yoke connection, where a yoke is attached to the frame at one end and to the wire at the other. The user may define the yoke length, width, weight, and connection point (Figure 3-10, Figure 3-11).

Anchor Flange Connection

The user can specify the connection point between the pipe and the frame (at the anchor flange) using a set of coordinates (Figure 3-10).

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A&R Connection Data:			
A&R Wire Attachment Type:	Yoke		
Length, L	7		
Width, W	2.082 m		
Weight in Air	3.0 Te		
Weight in Water	3.0 Te		
Connection Point (Relative to Frame Centroid)			
	X	Y	Z
	-0.098	-0.104	-0.758
Anchor Flange:			
Location (Relative to Frame Centroid)			
	X	Y	Z
	-3.017	0.0	-0.758

A&R Connection Data:			
A&R Wire Attachment Type:	Yoke		
Length, L	7		
Width, W	2.082 m		
Weight in Air	3.0 Te		
Weight in Water	3.0 Te		
Connection Point (Relative to Frame Centroid)			
	X	Y	Z
Frame Only	-0.098	-0.104	-0.758
Frame & Mudmat (Wings Up)	-0.098	-0.104	-1.22
Anchor Flange:			
Location (Relative to Frame Centroid)			
	X	Y	Z
Frame Only	-3.017	0.0	-0.758
Frame & Mudmat (Wings Up)	-4.283	0.0	-1.22

Figure 3-10 - Differences in A&R Connection and Anchor Flange Inputs between Modelling Types - Without Wings (Left) and With Wings (Right)

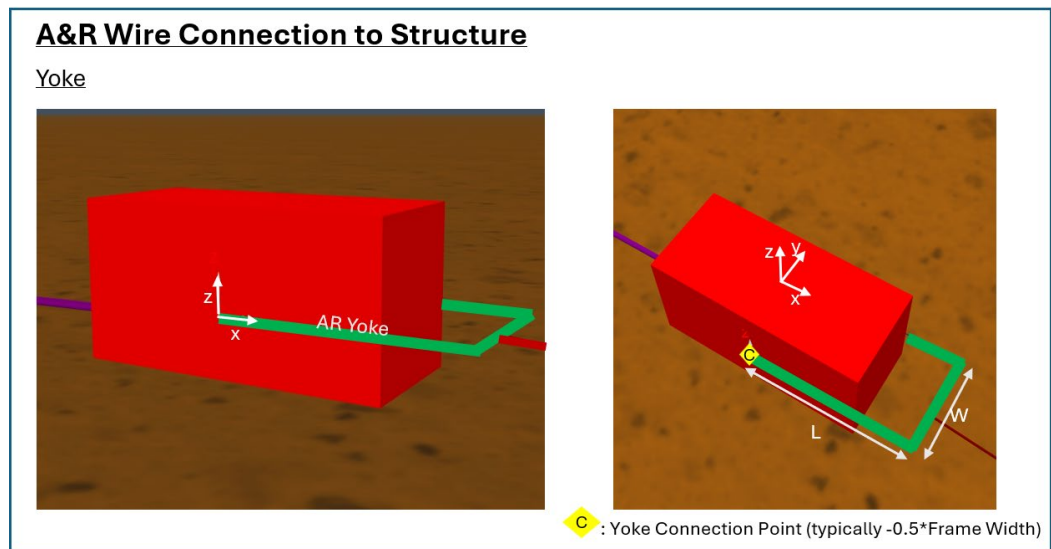


Figure 3-11 - A&R Wire Connection - Yoke Option

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3.2.5 Tab 5 – Buoyancy Input

The user can specify the following parameters for the buoyancy modules:

- Buoyancy connection type (Yoke or Running wire),
- Connection point to the frame (as in Figure 3-13),
- Length (Length of wire if using running wire, and length of the longest section of the yoke if using a yoke, as in Figure 3-13),
- Weight (if using yoke)
- Number of modules used (if more than one (1) is used, the modules will be arranged in series, as per Figure 3-14),
- Buoy uplift force [Te],
- Module dry weight [Te],
- Module length [m],
- Rigging OD [mm],
- Rigging length [m],
- Rigging mass [Te/m]

The screenshot shows the 'PG PLET Laydown Automation Tool' window with the 'Buoyancy Input' tab selected. The interface is divided into several sections:

- Use Buoyancy Modules?** (checked)
- Buoyancy Connection Data:**
 - Connection Type: Yoke (dropdown)
 - Connection Point (Relative to Frame Centroid):

	X	Y	Z
Frame Only	-0.395	-1.041	1.293
Frame & Mudmat (Wings Up)	-0.796	-1.041	0.831
- Buoyancy Yoke Data:**
 - Length (hinge to yoke padeye): 10 m
 - Weight in Air: 3 Te
 - Weight in Water: 3 Te
- Buoyancy Module Data:**
 - Number of Buoyancy Modules: 11 (dropdown)
 - Table:

	Uplift Force (Te)	Module Dry Weight	Module Length (m)	Rigging OD (mm)	Rigging Length (m)	Rigging Mass (Te/m)
1	10	5	2	32	5	0.004

The bottom of the window shows a tabbed interface with the following tabs: General / Environment, Pipe & Stress/Transition Joint Input, Code Check Input, Structure Input, Buoyancy Input (active), Sequence, Static Results, and Dynamic Screening.

Figure 3-12 - Buoyancy Options

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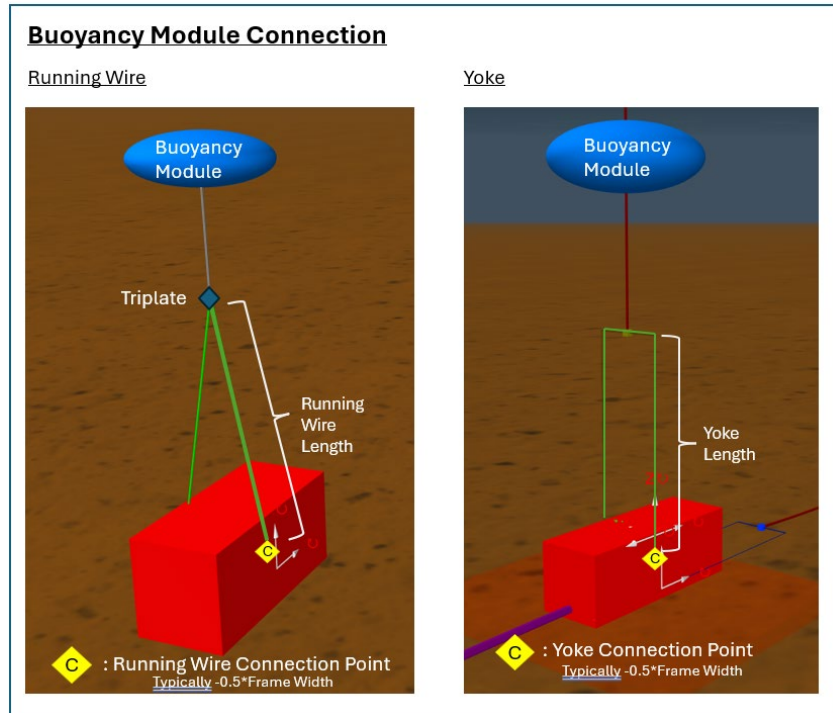


Figure 3-13 - Buoyancy Module Connection Types

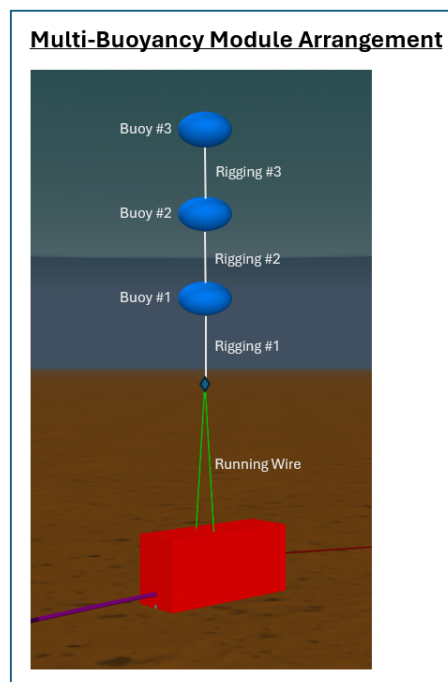


Figure 3-14 - Buoyancy Module Arrangement

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3.2.6 Tab 6 – Sequence Input

The sequence input tab takes in the vessel, deployment sequence, and pipe integrity limit inputs.

Figure 3-15 - Sequence Tab

The user may select between the three (3) reel lay vessels; Vega, Oceans, and Navica.

Seven Vega

- Set ramp angles are: [45.0, 49.4, 53.8, 58.2, 62.6, 67.1, 71.5, 76.1, 80.8, 85.6, 90.5, 95.0]. However, ramp angles outside of these values are also available due to the Vega ramp jacking system.
- AR Sheave locations: Port, starboard, port wire catcher, starboard wire catcher. Dual AR mode is also available with the use of an AR beam.
- Empty, mid, or full reel.

Seven Navica

- Set ramp angles are: [20.3, 24.1, 27.9, 31.8, 35.7, 39.7, 43.8, 47.9, 52.0, 56.3, 60.7, 65.1, 69.8, 74.5, 79.5, 84.6, 90.0]. Ramp angles outside of these options are not available.
- AR Sheave locations: Upper sheave and mid sheave.
- Full reel.

Seven Oceans

- Set ramp angles are: [24.3, 26.3, 28.3, 30.2, 32.1, 34.0, 35.9, 37.7, 39.6, 41.5, 43.3, 45.2, 47.1, 49.0, 50.9, 52.8, 54.7, 56.6, 58.5, 60.4, 62.4, 64.4,

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66.4, 68.4, 70.4, 72.5, 74.6, 76.7, 78.8, 81.0, 83.2, 85.4, 87.7, 90.0]. Ramp angles outside of these options are not available.

- AR Sheave locations: Upper sheave and mid sheave.
- Fully loaded or unloaded reel.

3.3 STEP 3: RUNNING THE CALCULATIONS

To run the calculations, the user will firstly need to save the input data.

Clicking on the highlighted *Save Current Data* button as in Figure 3-16 will prompt a new window whereby the user can specify the name and directory of the save file (and consequently the .dat files generated during the run), which will be in .json format.

To create the base files as specified in Section 2.1, the user may click the *Generate Basefile Only* button (highlighted in red in Figure 3-17). This allows the user to inspect the created files prior to running the full sequence.

To create the full static landing sequence, the user may click on the *Generate Laydown Sequence* button (highlighted in green in Figure 3-17).

A console window will update the user on the status of the simulation. An appearance of a table summarising the results of the sequence in the console window signifies completion of the program. Two (2) Excel files will appear in the user's selected save directory, as specified in Section 2.3, which contain the raw static results and the static laytable.

Figure 3-18 shows an example of how a working folder may look like on completion of a program, containing:

- the generated static .dat files,
- the json input file,
- the raw output data, and
- the laytable.

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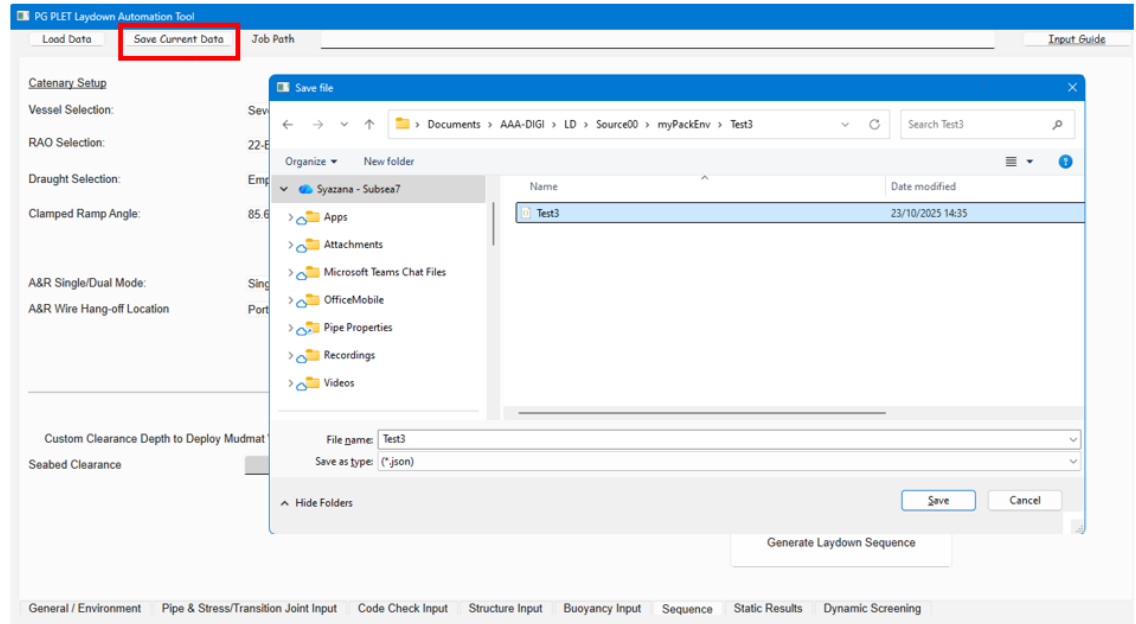


Figure 3-16 - Saving the Setup

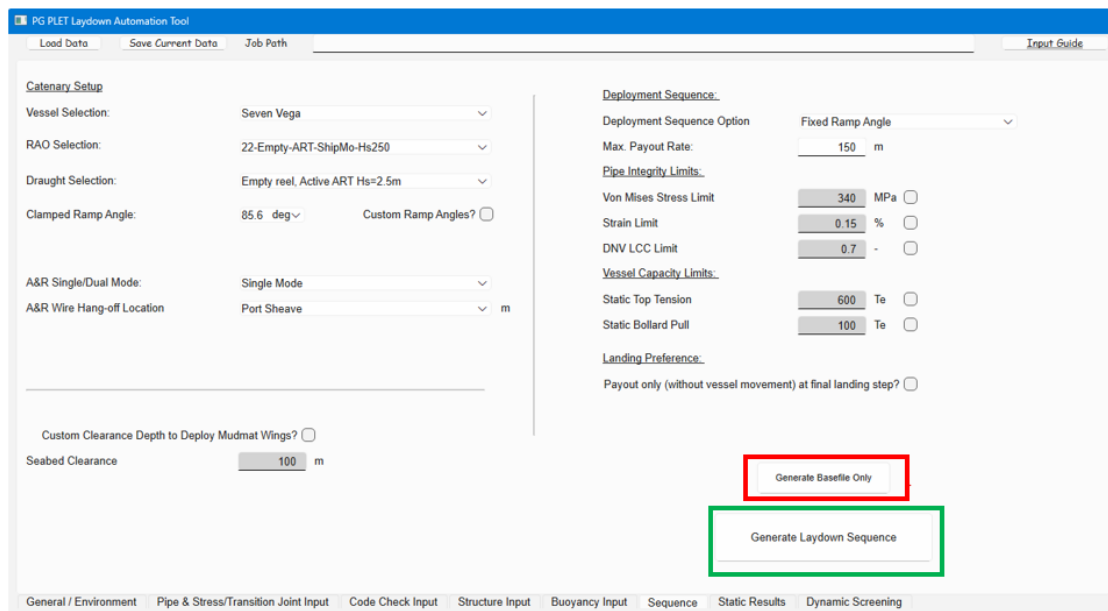


Figure 3-17 - Creating the Base Files (.dat) and the Sequence Steps (.dat)

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Name	Date modified	Type	Size
00_LD_76.1_Stage_1-BuoyYoke	23/10/2025 15:04	OrcaFlex binary data file	692 KB
01_LD_76.1_Stage_1	23/10/2025 15:04	OrcaFlex binary data file	699 KB
02_LD_76.1_Stage_1	23/10/2025 15:05	OrcaFlex binary data file	705 KB
03_LD_76.1_Stage_1	23/10/2025 15:05	OrcaFlex binary data file	709 KB
03_LD_76.1_Stage_1-SurfaceHydro	23/10/2025 15:05	OrcaFlex binary data file	813 KB
04_LD_76.1_Stage_1	23/10/2025 15:08	OrcaFlex binary data file	718 KB
05_LD_76.1_Stage_1	23/10/2025 15:16	OrcaFlex binary data file	727 KB
05_LD_76.1_Stage_1-Failed	23/10/2025 15:16	OrcaFlex binary data file	726 KB
06_LD_76.1_Stage_1	23/10/2025 15:19	OrcaFlex binary data file	726 KB
07_LD_76.1_Stage_1	23/10/2025 15:20	OrcaFlex binary data file	727 KB
08_LD_76.1_Stage_2	23/10/2025 15:20	OrcaFlex binary data file	727 KB
09_LD_76.1_Stage_2	23/10/2025 15:21	OrcaFlex binary data file	732 KB
010_LD_76.1_Stage_2	23/10/2025 15:22	OrcaFlex binary data file	731 KB
011_LD_76.1_Stage_2	23/10/2025 15:23	OrcaFlex binary data file	732 KB
012_LD_76.1_Stage_2	23/10/2025 15:23	OrcaFlex binary data file	734 KB
013_LD_76.1_Stage_2	23/10/2025 15:25	OrcaFlex binary data file	745 KB
Stage_2_PLET	23/10/2025 15:01	OrcaFlex binary data file	705 KB
Test2	22/10/2025 15:32	JSON Source File	8 KB
Test2_Laytable	23/10/2025 15:37	Microsoft Excel Worksheet	17 KB
Test2_StaticResults	23/10/2025 15:26	Microsoft Excel Worksheet	14 KB

Figure 3-18 - Example Output in the Save Directory

3.4 STEP 4: DISPLAYING THE RESULT SUMMARY IN THE GUI FOR DYNAMIC SCREENING SELECTION

To select the static steps to be sent for dynamic screening, the user can click the button highlighted in red in Figure 3-19. In the *Run Dynamic* column, the user can select which steps are to be sent for further analysis ie dynamic screening [HOLD].

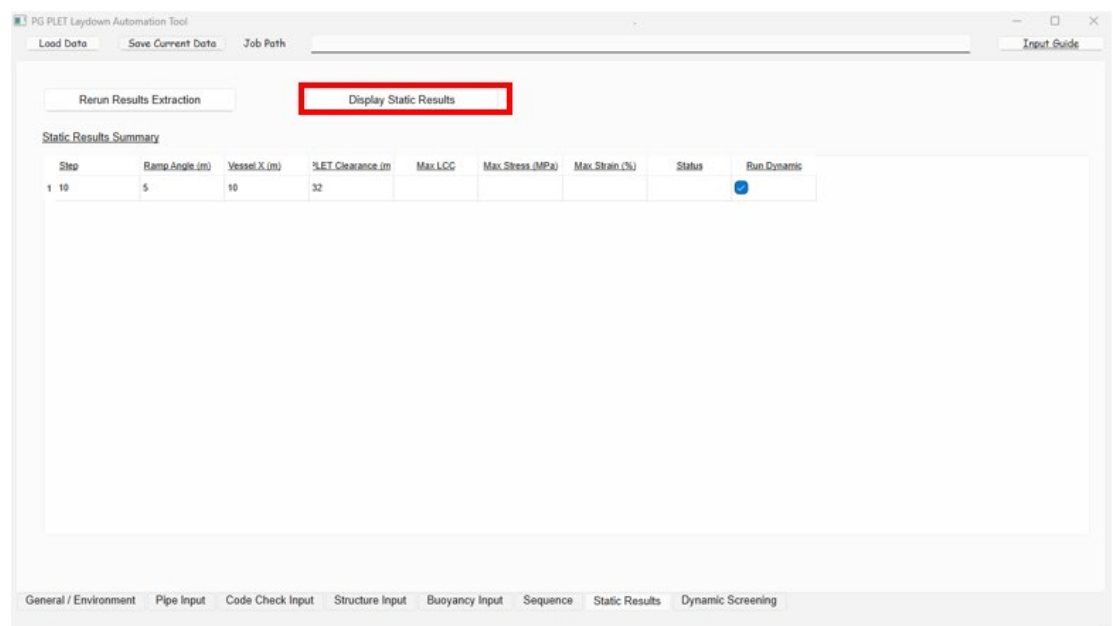


Figure 3-19 - Displaying a Result Summary and Selecting Steps for Dynamic Screening

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4. TROUBLESHOOTING

This section covers the common issues faced during testing.

4.1 STATIC CONVERGENCE

Whilst finding a static solution, OrcaFlex may not be able to achieve convergence. In event of non-convergence, the tool attempts the following:

- Increasing the damping factor,
- Increasing the static maximum number of iterations.

Modified parameters will be set back to default values for the next step once static convergence has been achieved.

4.2 CORRUPTED STATIC MODEL

It is possible that a model may converge towards an unphysical configuration, for instance if the pipe/wire/rigging has solved with a $>180^\circ$ bend, or the A&R beam has solved upside down.

In event of this occurrence, it is possible for the user to manually adjust the affected step's .dat file, save it, and then rerun the result extraction by pressing the *Rerun Results Extraction* button.

Note that by doing this, the previously generated raw results file and laytable will be overwritten.

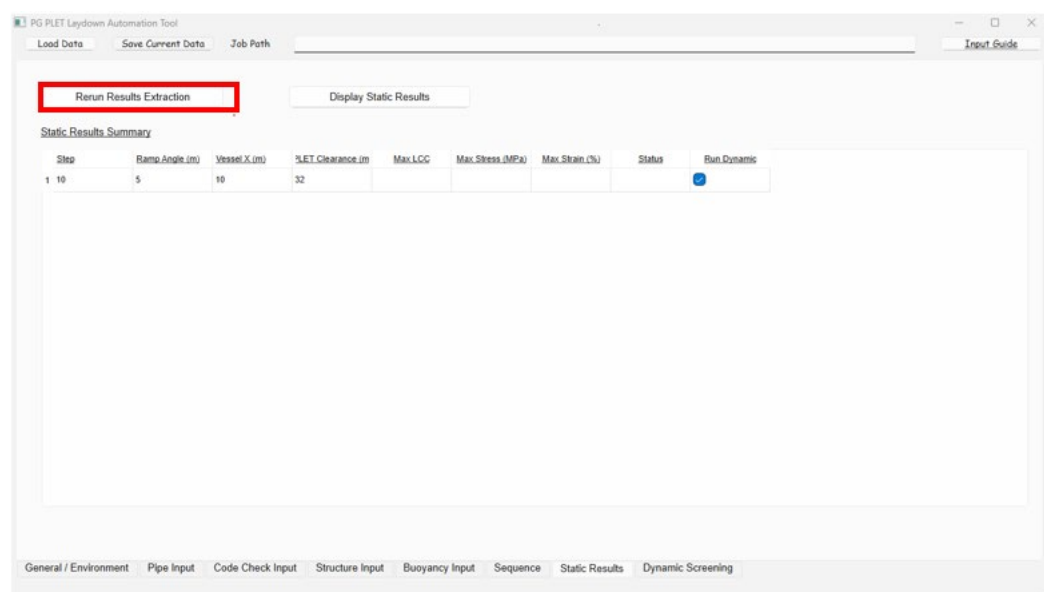


Figure 4-1 - Rerun Results Extraction